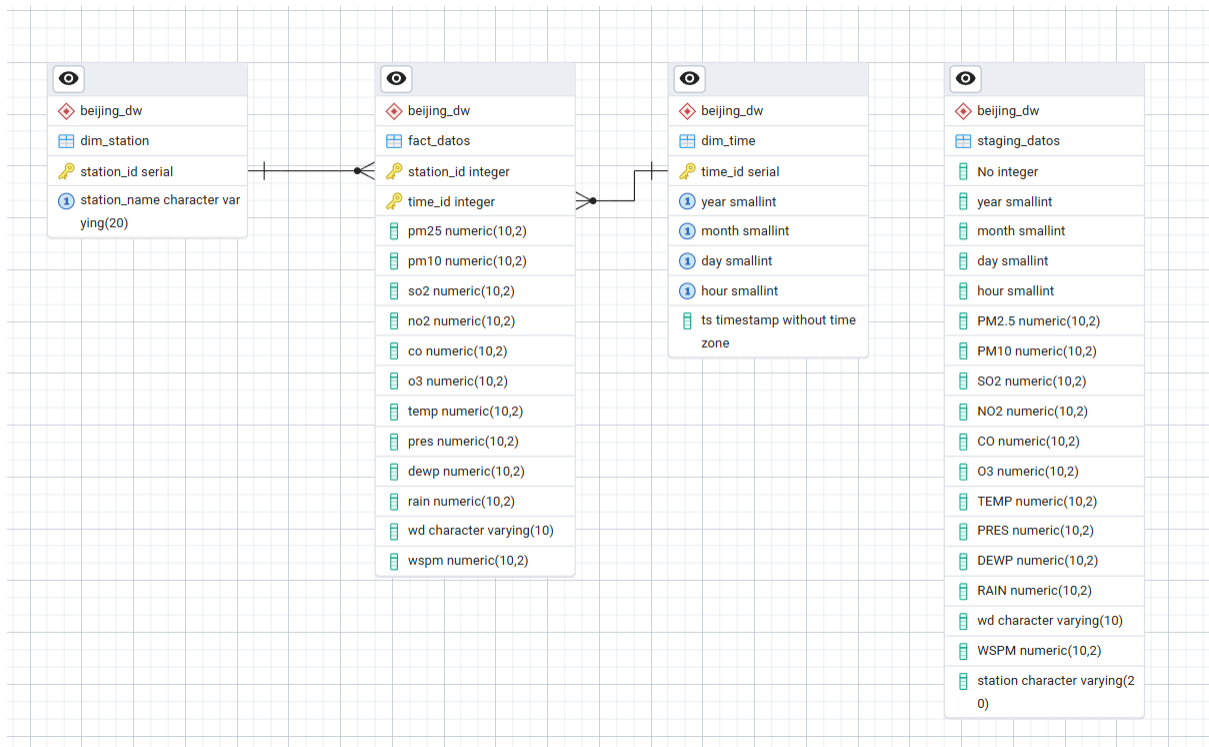


Project Beijing

Equipo Rojo



CREACIÓN DE SCHEMA, TABLAS E INSERCIÓN DE DATOS

```
CREATE SCHEMA beijing_dw;

CREATE TABLE beijing_dw.dim_station (
    station_id SERIAL PRIMARY KEY,
    station_name VARCHAR(20) UNIQUE NOT NULL
);

CREATE TABLE beijing_dw.dim_time (
    time_id SERIAL PRIMARY KEY,
    year SMALLINT NOT NULL,
    month SMALLINT NOT NULL,
    day SMALLINT NOT NULL,
    hour SMALLINT NOT NULL,
```

```

    ts TIMESTAMP NOT NULL,
    UNIQUE (year, month, day, hour)
);

CREATE TABLE beijing_dw.fact_datos (
    station_id INT NOT NULL,
    time_id INT NOT NULL,

    pm25 NUMERIC(10,2),
    pm10 NUMERIC(10,2),
    so2 NUMERIC(10,2),
    no2 NUMERIC(10,2),
    co NUMERIC(10,2),
    o3 NUMERIC(10,2),
    temp NUMERIC(10,2),
    pres NUMERIC(10,2),
    dewp NUMERIC(10,2),
    rain NUMERIC(10,2),
    wd VARCHAR(10),
    wspm NUMERIC(10,2),

    PRIMARY KEY (station_id, time_id),

    CONSTRAINT fk_station
        FOREIGN KEY (station_id)
        REFERENCES beijing_dw.dim_station(station_id),

    CONSTRAINT fk_time
        FOREIGN KEY (time_id)
        REFERENCES beijing_dw.dim_time(time_id)
);

CREATE TABLE beijing_dw.staging_datos (
    "No" INT,
    "year" SMALLINT,
    "month" SMALLINT,
    "day" SMALLINT,
    "hour" SMALLINT,
    "PM2.5" NUMERIC(10,2),
    "PM10" NUMERIC(10,2),
    "SO2" NUMERIC(10,2),
    "NO2" NUMERIC(10,2),

```

```

"CO" NUMERIC(10,2),
"O3" NUMERIC(10,2),
"TEMP" NUMERIC(10,2),
"PRES" NUMERIC(10,2),
"DEWP" NUMERIC(10,2),
"RAIN" NUMERIC(10,2),
"wd" VARCHAR(10),
"WSPM" NUMERIC(10,2),
"station" VARCHAR(20)
);

-- Aotizhongxin
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Aotizhongxin_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Changping
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Changping_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Dingling
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Dingling_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Dongsi
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-

```

```
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Dongsi_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Guanyuan
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Guanyuan_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Gucheng
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Gucheng_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Huairou
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Huairou_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Nongzhanguan
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Nongzhanguan_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';
```

```

-- Shunyi
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Shunyi_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Tiantan
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Tiantan_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Wanliu
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Wanliu_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

-- Wanshouxigong
COPY beijing_dw.staging_datos
FROM 'C:\beijing+multi+site+air+quality+data\PRSA2017_Data_20130301-
20170228\PRSA_Data_20130301-20170228\PRSA_Data_Wanshouxigong_20130301-
20170228.csv'
DELIMITER ','
CSV HEADER
NULL 'NA';

INSERT INTO beijing_dw.dim_station (station_name)
SELECT DISTINCT station
FROM beijing_dw.staging_datos;

INSERT INTO beijing_dw.dim_time (year, month, day, hour, ts)
SELECT DISTINCT

```

```

    year,
    month,
    day,
    hour,
    make_timestamp(year, month, day, hour, 0, 0)
FROM beijing_dw.staging_datos;

INSERT INTO beijing_dw.fact_datos (
    station_id,
    time_id,
    pm25, pm10, so2, no2, co, o3,
    temp, pres, dewp, rain, wd, wspm
)
SELECT
    s.station_id,
    t.time_id,
    d."PM2.5",
    d."PM10",
    d."SO2",
    d."NO2",
    d."CO",
    d."O3",
    d."TEMP",
    d."PRES",
    d."DEWP",
    d."RAIN",
    d.wd,
    d."WSPM"
FROM beijing_dw.staging_datos d
JOIN beijing_dw.dim_station s
    ON s.station_name = d.station
JOIN beijing_dw.dim_time t
    ON (t.year, t.month, t.day, t.hour) =
        (d.year, d.month, d.day, d.hour);

CREATE INDEX idx_fact_station ON beijing_dw.fact_datos (station_id);
CREATE INDEX idx_fact_time ON beijing_dw.fact_datos (time_id);
CREATE INDEX idx_time_ts ON beijing_dw.dim_time (ts);

```

TABLA 1. OBSERVACIÓN DE LOS DATOS

```
-- Obtener PM2,5
WITH diario_consolidado AS (
SELECT
    s.station_name AS station,
    t.year,
    t.month,
    t.day,
    ROUND(AVG(f.pm25), 2) AS pm25_diario,
    ROUND(AVG(f.pm10), 2) AS pm10_diario,
    ROUND(AVG(f.so2), 2) AS so2_diario,
    ROUND(AVG(f.no2), 2) AS no2_diario,
    ROUND(AVG(f.co/1000), 2) AS co_diario,
    ROUND(AVG(f.o3), 2) AS o3_diario
FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
GROUP BY s.station_name, t.year, t.month, t.day
)
SELECT
    AVG(pm25_diario) AS avg_pm25,
    COUNT(pm25_diario) AS count_pm25,
    MAX(pm25_diario) AS max_pm25,
    MIN(pm25_diario) AS min_pm25,
    SUM(CASE WHEN pm25_diario IS NULL THEN 1 ELSE 0 END) AS null_count,
    stddev(pm25_diario) AS variacion_pm25
FROM diario_consolidado;

-- Obtener pm10
WITH diario_consolidado AS (
SELECT
    s.station_name AS station,
    t.year,
    t.month,
    t.day,
    ROUND(AVG(f.pm25), 2) AS pm25_diario,
    ROUND(AVG(f.pm10), 2) AS pm10_diario,
    ROUND(AVG(f.so2), 2) AS so2_diario,
    ROUND(AVG(f.no2), 2) AS no2_diario,
    ROUND(AVG(f.co/1000), 2) AS co_diario,
    ROUND(AVG(f.o3), 2) AS o3_diario
FROM beijing_dw.fact_datos f
```

```

JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
GROUP BY s.station_name, t.year, t.month, t.day
)
SELECT
    AVG(pm10_diario) AS avg_pm10,
    COUNT(pm10_diario) AS count_pm10,
    MAX(pm10_diario) AS max_pm10,
    MIN(pm10_diario) AS min_pm10,
    SUM(CASE WHEN pm10_diario IS NULL THEN 1 ELSE 0 END) AS null_count,
    stddev(pm10_diario) AS variacion_pm10
FROM diario_consolidado;

-- Obtener promedios so2 (Corregido según orden original)
WITH diario_consolidado AS (
SELECT
    s.station_name AS station,
    t.year,
    t.month,
    t.day,
    ROUND(AVG(f.pm25), 2) as pm25_diario,
    ROUND(AVG(f.pm10), 2) as pm10_diario,
    ROUND(AVG(f.so2), 2) as so2_diario,
    ROUND(AVG(f.no2), 2) as no2_diario,
    ROUND(AVG(f.co/1000), 2) as co_diario,
    ROUND(AVG(f.o3), 2) as o3_diario
FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
GROUP BY s.station_name, t.year, t.month, t.day
)
SELECT
    AVG(so2_diario) AS avg_so2,
    COUNT(so2_diario) AS count_so2,
    MAX(so2_diario) AS max_so2,
    MIN(so2_diario) AS min_so2,
    SUM(CASE WHEN so2_diario IS NULL THEN 1 ELSE 0 END) AS null_count,
    stddev(so2_diario) AS variacion_so2
FROM diario_consolidado;

-- Obtener promedios no2
WITH diario_consolidado AS (

```

```

SELECT
    s.station_name AS station,
    t.year,
    t.month,
    t.day,
    ROUND(AVG(f.pm25), 2) AS pm25_diario,
    ROUND(AVG(f.pm10), 2) AS pm10_diario,
    ROUND(AVG(f.so2), 2) AS so2_diario,
    ROUND(AVG(f.no2), 2) AS no2_diario,
    ROUND(AVG(f.co/1000), 2) AS co_diario,
    ROUND(AVG(f.o3), 2) AS o3_diario
FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
GROUP BY s.station_name, t.year, t.month, t.day
)
SELECT
    AVG(no2_diario) AS avg_no2,
    COUNT(no2_diario) AS count_no2,
    MAX(no2_diario) AS max_no2,
    MIN(no2_diario) AS min_no2,
    SUM(CASE WHEN no2_diario IS NULL THEN 1 ELSE 0 END) AS null_count,
    stddev(no2_diario) AS variacion_no2
FROM diario_consolidado;

-- Obtener promedios o3
WITH o3_movil_8h AS (
    SELECT
        f.station_id,
        t.year,
        t.month,
        t.day,
        t.hour,
        f.o3,
        ROUND(
            AVG(f.o3) OVER (
                PARTITION BY f.station_id
                ORDER BY t.year, t.month, t.day, t.hour
                ROWS BETWEEN 7 PRECEDING AND CURRENT ROW
            ), 2
        ) AS o3_promedio_8h
    FROM beijing_dw.fact_datos f

```

```

        JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
        WHERE f.o3 IS NOT NULL
    ),
    o3_diario_max AS (
        SELECT
            station_id,
            year,
            month,
            day,
            ROUND(MAX(o3_promedio_8h), 2) AS o3_max_8h_diario
        FROM o3_movil_8h
        GROUP BY station_id, year, month, day
    )
SELECT
    AVG(o3_max_8h_diario) AS avg_o3,
    COUNT(o3_max_8h_diario) AS count_o3,
    MAX(o3_max_8h_diario) AS max_o3,
    MIN(o3_max_8h_diario) AS min_o3,
    SUM(CASE WHEN o3_max_8h_diario IS NULL THEN 1 ELSE 0 END) AS
null_count,
    stddev(o3_max_8h_diario) AS variacion_o3
FROM o3_diario_max;

-- Obtener promedios de CO
WITH diario_consolidado AS (
    SELECT
        s.station_name AS station,
        t.year,
        t.month,
        t.day,
        ROUND(AVG(f.pm25), 2) AS pm25_diario,
        ROUND(AVG(f.pm10), 2) AS pm10_diario,
        ROUND(AVG(f.so2), 2) AS so2_diario,
        ROUND(AVG(f.no2), 2) AS no2_diario,
        ROUND(AVG(f.co/1000), 2) AS co_diario,
        ROUND(AVG(f.o3), 2) AS o3_diario
    FROM beijing_dw.fact_datos f
    JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
    JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
    GROUP BY s.station_name, t.year, t.month, t.day
)
SELECT

```

```

AVG(co_diario) AS avg_co,
COUNT(co_diario) AS count_co,
MAX(co_diario) AS max_co,
MIN(co_diario) AS min_co,
SUM(CASE WHEN co_diario IS NULL THEN 1 ELSE 0 END) AS null_count,
stddev(co_diario) AS variacion_co
FROM diario_consolidado;

```

TABLA 2. IDENTIFICACIÓN DE LAS 5 ESTACIONES CON MAYOR PROMEDIO DE PM2.5.

```

WITH diario_consolidado AS (
SELECT
    s.station_name AS station,
    t.year,
    t.month,
    t.day,
    ROUND(AVG(f.pm25), 2) AS pm25_diario,
    ROUND(AVG(f.pm10), 2) AS pm10_diario,
    ROUND(AVG(f.temp), 2) AS temp_promedio,
    SUM(f.rain) AS lluvia_total
FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
GROUP BY s.station_name, t.year, t.month, t.day
ORDER BY s.station_name, t.year, t.month, t.day
)
SELECT
    station,
    CASE
        WHEN station IN ('Dongsi', 'Guanyuan', 'Aotizhongxin',
'Tiantan', 'Wanshouxigong', 'Nongzhanguan') THEN 'URBANA'
        WHEN station IN ('Dingling', 'Huairou', 'Changping') THEN
'EXTERIOR'
        WHEN station IN ('Gucheng', 'Wanliu', 'Shunyi') THEN 'SUBURBANA'
    END AS ubicacion,
    AVG(pm25_diario) AS ave_pm25,
    COUNT(pm25_diario) AS count_pm25,
    MAX(pm25_diario) AS max_pm25,
    MIN(pm25_diario) AS min_pm25,
    SUM(CASE WHEN pm25_diario IS NULL THEN 1 ELSE 0 END) AS null_count,

```

```

stddev(pm25_diario) AS variacion_pm25
FROM diario_consolidado
GROUP BY station
ORDER BY AVG(pm25_diario) DESC
LIMIT 5;

```

TABLA 3. IDENTIFICACIÓN DE LAS 5 ESTACIONES CON MEJOR PROMEDIO DE PM2.5.

```

WITH diario_consolidado AS (
SELECT
    s.station_name AS station,
    t.year,
    t.month,
    t.day,
    ROUND(AVG(f.pm25), 2) AS pm25_diario,
    ROUND(AVG(f.pm10), 2) AS pm10_diario,
    ROUND(AVG(f.temp), 2) AS temp_promedio,
    SUM(f.rain) AS lluvia_total
FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
GROUP BY s.station_name, t.year, t.month, t.day
ORDER BY s.station_name, t.year, t.month, t.day
)

SELECT
    station,
    CASE
        WHEN station IN ('Dongsi', 'Guanyuan', 'Aotizhongxin',
'Tiantan', 'Wanshouxigong', 'Nongzhanguan') THEN 'URBANA'
        WHEN station IN ('Dingling', 'Huairou', 'Changping') THEN
'EXTERIOR'
        WHEN station IN ('Gucheng', 'Wanliu', 'Shunyi') THEN 'SUBURBANA'
    END AS ubicacion,
    AVG(pm25_diario) AS ave_pm25,
    COUNT(pm25_diario) AS count_pm25,
    MAX(pm25_diario) AS max_pm25,
    MIN(pm25_diario) AS min_pm25,
    SUM(CASE WHEN pm25_diario IS NULL THEN 1 ELSE 0 END) AS null_count,
    stddev(pm25_diario) AS variacion_pm25

```

```
FROM diario_consolidado
GROUP BY station
ORDER BY AVG(pm25_diario) ASC
LIMIT 5;
```

TABLA 4. OBSERVACIÓN DE LAS VARIABLES DEL ENTORNO POR ZONA

```
WITH diario_consolidado AS (
    SELECT
        s.station_name AS station,
        t.year,
        t.month,
        t.day,
        ROUND(AVG(f.pm25), 2) AS pm25_diario,
        ROUND(AVG(f.pm10), 2) AS pm10_diario,
        ROUND(AVG(f.temp), 2) AS temp_promedio,
        ROUND(AVG(f.wspm), 2) AS wind_speed,
        ROUND(AVG(f.pres), 2) AS press_prom,
        ROUND(AVG(f.so2), 2) AS so2_prom,
        ROUND(AVG(f.no2), 2) AS no2_prom,
        SUM(f.rain) AS lluvia_total,
        CASE
            WHEN s.station_name IN ('Dongsi', 'Guanyuan',
                'Aotizhongxin', 'Tiantan', 'Wanshouxigong', 'Nongzhanguan') THEN
                'URBANA'
            WHEN s.station_name IN ('Dingling', 'Huairou', 'Changping')
            THEN 'EXTERIOR'
            WHEN s.station_name IN ('Gucheng', 'Wanliu', 'Shunyi') THEN
                'SUBURBANA'
        END as ubicacion
    FROM beijing_dw.fact_datos f
    JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
    JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
    GROUP BY s.station_name, t.year, t.month, t.day
)

SELECT
    ubicacion,
    ROUND(AVG(pm25_diario), 2) AS avg_pm25,
    ROUND(AVG(wind_speed), 2) AS wspeed_ubicacion,
```

```

ROUND(AVG(so2_prom), 2) AS so2_ubicacion,
ROUND(AVG(no2_prom), 2) AS no2_ubicacion,
ROUND(AVG(lluvia_total), 2) AS lluviaprom_ubicacion
FROM diario_consolidado
GROUP BY ubicacion
ORDER BY ubicacion, avg_pm25 ASC;

```

TABLA 5. OBSERVACIÓN DE LAS VARIABLES DEL ENTORNO POR ESTACIONES FUERA DE RANGOS NORMALES.

```

WITH diario_consolidado AS (
    SELECT
        s.station_name AS station,
        t.year,
        t.month,
        t.day,
        ROUND(AVG(f.pm25), 2) AS pm25_diario,
        ROUND(AVG(f.pm10), 2) AS pm10_diario,
        ROUND(AVG(f.temp), 2) AS temp_promedio,
        ROUND(AVG(f.wspm), 2) AS wind_speed,
        ROUND(AVG(f.pres), 2) AS press_prom,
        ROUND(AVG(f.so2), 2) AS so2_prom,
        ROUND(AVG(f.no2), 2) AS no2_prom,
        SUM(f.rain) AS lluvia_total,
        CASE
            WHEN s.station_name IN ('Dongsi', 'Guanyuan',
'Aotizhongxin', 'Tiantan', 'Wanshouxigong', 'Nongzhanguan') THEN
'URBANA'
            WHEN s.station_name IN ('Dingling', 'Huairou', 'Changping')
THEN 'EXTERIOR'
            WHEN s.station_name IN ('Gucheng', 'Wanliu', 'Shunyi') THEN
'SUBURBANA'
        END as ubicacion
    FROM beijing_dw.fact_datos f
    JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
    JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
    GROUP BY s.station_name, t.year, t.month, t.day
)

SELECT

```

```

station,
CASE
    WHEN station IN ('Dongsi', 'Guanyuan', 'Aotizhongxin',
'Tiantan', 'Wanshouxigong', 'Nongzhanguan') THEN 'URBANA'
    WHEN station IN ('Dingling', 'Huairou', 'Changping') THEN
'EXTERIOR'
    WHEN station IN ('Gucheng', 'Wanliu', 'Shunyi') THEN 'SUBURBANA'
END as ubicacion,
ROUND(AVG(pm25_diario), 2) AS avg_pm25,
ROUND(AVG(wind_speed), 2) AS wspeed_ubicacion,
ROUND(AVG(so2_prom), 2) AS so2_ubicacion,
ROUND(AVG(no2_prom), 2) AS no2_ubicacion,
ROUND(AVG(lluvia_total), 2) AS lluviaprom_ubicacion
FROM diario_consolidado
WHERE station IN ('Tiantan', 'Wanliu', 'Gucheng')
GROUP BY station
ORDER BY ubicacion, avg_pm25;

```

TABLA 6. OBSERVACIÓN DEL COMPORTAMIENTO DEL PM2.5 EN LAS DIFERENTES ESTACIONES.

```

WITH diario_consolidado AS (
    SELECT
        s.station_name AS station,
        t.year,
        t.month,
        t.day,
        ROUND(AVG(f.pm25), 2) AS pm25_diario,
        ROUND(AVG(f.pm10), 2) AS pm10_diario,
        ROUND(AVG(f.temp), 2) AS temp_promedio,
        SUM(f.rain) AS lluvia_total,
        CASE
            WHEN t.month IN (12, 1, 2) THEN 'Winter'
            WHEN t.month IN (3, 4, 5) THEN 'Spring'
            WHEN t.month IN (6, 7, 8) THEN 'Summer'
            WHEN t.month IN (9, 10, 11) THEN 'Autumn'
        END AS season
    FROM beijing_dw.fact_datos f
    JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
    JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
    GROUP BY s.station_name, t.year, t.month, t.day

```

```
)

SELECT
    season,
    ROUND(AVG(pm25_diario), 2) AS avg_pm25,
    ROUND(COUNT(pm25_diario), 2) AS count_pm25,
    ROUND(MAX(pm25_diario), 2) AS max_pm25,
    ROUND(MIN(pm25_diario), 2) AS min_pm25,
    SUM(CASE WHEN pm25_diario IS NULL THEN 1 ELSE 0 END) AS null_count,
    STDDEV(pm25_diario) AS variacion_pm25
FROM diario_consolidado
GROUP BY season;
```

TABLA 7. COMPORTAMIENTO DEL NO2 Y SO2 FRENTE AL PM2.5.

```
WITH diario_consolidado AS (
    SELECT
        s.station_name AS station,
        t.year,
        t.month,
        t.day,
        ROUND(AVG(f.pm25), 2) AS pm25_diario,
        ROUND(AVG(f.pm10), 2) AS pm10_diario,
        ROUND(AVG(f.temp), 2) AS temp_promedio,
        ROUND(AVG(f.wspm), 2) AS wind_speed,
        ROUND(AVG(f.pres), 2) AS press_prom,
        ROUND(AVG(f.so2), 2) AS so2_prom,
        ROUND(AVG(f.no2), 2) AS no2_prom,
        SUM(f.rain) AS lluvia_total,
        CASE
            WHEN t.month IN (12, 1, 2) THEN 'Invierno'
            WHEN t.month IN (3, 4, 5) THEN 'Primavera'
            WHEN t.month IN (6, 7, 8) THEN 'Verano'
            WHEN t.month IN (9, 10, 11) THEN 'Otoño'
        END AS season
    FROM beijing_dw.fact_datos f
    JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
    JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
    GROUP BY s.station_name, t.year, t.month, t.day
)
```

```

SELECT
    season,
    ROUND(AVG(pm25_diario), 2) AS avg_pm25,
    ROUND(AVG(so2_prom), 2) AS so2_season,
    ROUND(AVG(no2_prom), 2) AS no2_season
FROM diario_consolidado
GROUP BY season
ORDER BY season, avg_pm25;

```

TABLA 8. COMPORTAMIENTO DE FACTORES CLIMATOLÓGICOS.

```

WITH diario_consolidado AS (
    SELECT
        s.station_name AS station,
        t.year,
        t.month,
        t.day,
        ROUND(AVG(f.pm25), 2) AS pm25_diario,
        ROUND(AVG(f.pm10), 2) AS pm10_diario,
        ROUND(AVG(f.temp), 2) AS temp_promedio,
        ROUND(AVG(f.wspm), 2) AS wind_speed,
        ROUND(AVG(f.pres), 2) AS press_prom,
        ROUND(AVG(f.so2), 2) AS so2_prom,
        ROUND(AVG(f.no2), 2) AS no2_prom,
        SUM(f.rain) AS lluvia_total,
        CASE
            WHEN t.month IN (12, 1, 2) THEN 'Invierno'
            WHEN t.month IN (3, 4, 5) THEN 'Primavera'
            WHEN t.month IN (6, 7, 8) THEN 'Verano'
            WHEN t.month IN (9, 10, 11) THEN 'Otoño'
        END AS season
    FROM beijing_dw.fact_datos f
    JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
    JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
    GROUP BY s.station_name, t.year, t.month, t.day
)

SELECT
    season,

```

```

ROUND(AVG(pm25_diario), 2) AS avg_pm25,
ROUND(AVG(lluvia_total), 2) AS lluvia_season,
ROUND(AVG(wind_speed), 2) AS wind_season
FROM diario_consolidado
GROUP BY season
ORDER BY season, avg_pm25;

```

TABLA 9. EL VIENTO COMO FACTOR CLIMATOLÓGICO.

```

SELECT
    s.station_name AS station,
    ROUND(AVG(f.wspm), 2) AS velocidad_viento,
    ROUND(AVG(f.pm25), 2) AS media_pm25,
    ROUND(AVG(f.pm10), 2) AS media_pm10,
    ROUND(AVG(f.pm25) / NULLIF(AVG(f.wspm),0), 2) AS
estancamiento_pm25,
    ROUND(AVG(f.pm10) / NULLIF(AVG(f.wspm),0), 2) AS
estancamiento_pm10
FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
GROUP BY s.station_name
ORDER BY estancamiento_pm25 DESC;

```

TABLA 10. COMPARACIÓN DE PM2.5 ENTRE EL DÍA Y LA NOCHE

```

SELECT
    s.station_name AS station,
    ROUND(AVG(CASE WHEN t.hour BETWEEN 7 AND 18 THEN f.pm25
END)::numeric, 2) AS day,
    ROUND(AVG(CASE WHEN t.hour NOT BETWEEN 7 AND 18 THEN f.pm25
END)::numeric, 2) AS night
FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
GROUP BY s.station_name
ORDER BY s.station_name;

```

TABLA 11. LA LLUVIA COMO FACTOR CLIMATOLÓGICO. DETECCIÓN DE EPISODIOS (>150 $\mu\text{g}/\text{m}^3$)

```
WITH flagged AS (  
  SELECT  
    s.station_name AS station,  
    t.ts,  
    f.pm25,  
    (f.pm25 > 150) AS exceed  
  FROM beijing_dw.fact_datos f  
  JOIN beijing_dw.dim_station s ON f.station_id = s.station_id  
  JOIN beijing_dw.dim_time t ON f.time_id = t.time_id  
) ,  
islands AS (  
  SELECT *,  
    ts - ROW_NUMBER() OVER (PARTITION BY station ORDER BY ts) *  
  INTERVAL '1 hour' AS grp  
  FROM flagged  
  WHERE exceed  
)  
SELECT  
  station,  
  MIN(ts) AS episode_start,  
  MAX(ts) AS episode_end,  
  COUNT(*) AS hours_duration,  
  MAX(pm25) AS peak_pm25,  
  ROUND(AVG(pm25), 2) AS avg_pm25  
FROM islands  
GROUP BY station, grp  
HAVING COUNT(*) >= 6  
ORDER BY hours_duration DESC, peak_pm25 DESC;  
  
WITH lluvia_ctx AS (  
  SELECT  
    s.station_name AS station,  
    t.ts,  
    f.rain,  
    f.pm10,  
  
    -- antes  
    LAG(f.pm10, 1) OVER (  

```

```

        PARTITION BY s.station_name
        ORDER BY t.ts
    ) AS pm10_before,

    -- después (corto plazo)
    LEAD(f.pm10, 1) OVER (
        PARTITION BY s.station_name
        ORDER BY t.ts
    ) AS pm10_after_1h,

    -- medio plazo
    LEAD(f.pm10, 72) OVER (
        PARTITION BY s.station_name
        ORDER BY t.ts
    ) AS pm10_after_3d,

    LEAD(f.pm10, 120) OVER (
        PARTITION BY s.station_name
        ORDER BY t.ts
    ) AS pm10_after_5d,

    LEAD(f.pm10, 168) OVER (
        PARTITION BY s.station_name
        ORDER BY t.ts
    ) AS pm10_after_7d

FROM beijing_dw.fact_datos f
JOIN beijing_dw.dim_station s ON f.station_id = s.station_id
JOIN beijing_dw.dim_time t ON f.time_id = t.time_id
WHERE f.pm10 IS NOT NULL
)
SELECT
    station,
    ROUND(AVG(pm10_before), 2) AS pm10_before_rain,
    ROUND(AVG(pm10), 2) AS pm10_during_rain,
    ROUND(AVG(pm10_after_1h), 2) AS pm10_after_1h,
    ROUND(AVG(pm10_after_3d), 2) AS pm10_after_3d,
    ROUND(AVG(pm10_after_5d), 2) AS pm10_after_5d,
    ROUND(AVG(pm10_after_7d), 2) AS pm10_after_7d
FROM lluvia_ctx
WHERE rain > 0
GROUP BY station

```

```
ORDER BY station;
```